

Saptarshi Sadhu

Curriculum Vitae

Department of CSE, Adamas University
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Professional Summary

Overview Computer Science undergraduate specializing in Artificial Intelligence & Machine Learning at Adamas University, with research interests spanning healthcare AI, Edge AI, brain research & neurotech, intelligent systems, and environmental data science. Experienced in developing machine learning pipelines for large-scale real-world datasets, building scalable software systems, and leading large-scale open-source engineering initiatives. Passionate about advancing reliable, scalable, and impactful intelligent systems for healthcare and environmental applications through the integration of research and software engineering.

Research Interests

Healthcare & Bio-AI AI for Asthma & Respiratory Health, Brain Research, Neurotech & Depression Mitigation

Machine Learning Artificial Intelligence, Machine Learning, Edge AI & TinyML, Explainable AI (XAI)

Systems Distributed Systems, Cyber-Physical Systems, Intelligent Systems

Data Science Environmental Data Science, Time Series Analysis & Forecasting, Geospatial AI

Education

2024–2028 Bachelor of Technology in Computer Science & Engineering
Specialization: Artificial Intelligence & Machine Learning
Adamas University, Kolkata, India.
CGPA: 9.30 / 10.00
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Probability & Statistics, Linear Algebra.

2019–2021 Higher Secondary Education (Class XII)
B.T. Road Govt. Spon. H.S. School
Kolkata, India.
Score: 89.6% · **Stream:** Pure Science (Physics, Chemistry, Mathematics, Biology – PCMB)

2017–2019 Secondary Education (Class X)
B.T. Road Govt. Spon. H.S. School
Kolkata, India.
Score: 93.7%

Research Experience

Aug 2025 – Nov
2025

Lead ML Researcher

Urban Air Quality Analytics

Adamas University, Kolkata, India.

- **Dataset & Preprocessing:** Curated continuous ambient air quality monitoring (CAAQMS) datasets across 13 stations; engineered pipeline with timestamp alignment, missing value imputation, and Hampel / IQR outlier filtering.
- **ML Benchmarking:** Evaluated Linear Regression, Decision Trees, SVR, XGBoost, and LSTMs; achieved optimal non-linear predictive mapping with Random Forest modeling VOC, NO_x, and O₃ concentrations.
- **Atmospheric Dynamics:** Analyzed diurnal trends, seasonal heatmaps, and ozone-titration photochemical dynamics under meteorological parameters.
- **Publication:** Co-authored research manuscripts under preparation for journal submission.

Publications & Manuscripts

Manuscript in
Preparation (2026)

Argha Kamal Guha, **Saptarshi Sadhu**, Chirag Deb. "*Ozone Photochemical Regimes and Particulate Transport in Kolkata and Howrah.*" Manuscript in preparation, 2026.

Manuscript in
Preparation (2026)

Argha Kamal Guha, **Saptarshi Sadhu**, Susmit Bagchi, Chirag Deb. "*A Statistical Distribution, Spatial Autocorrelation, and Recurrent Neural Network Framework for High-Resolution Air Quality Forecasting in the Kolkata-Howrah Metropolitan Airshed, India.*" Manuscript in preparation, 2026.

Manuscript in
Preparation (2026)

Argha Kamal Guha, Enubothula Bala Mani, Piyush Raj, Navnit Kumar, **Saptarshi Sadhu**, Snehamanju Basu. "*Assessment of Particulate and Gaseous Pollution and its Statistical Distribution: A Case Study of Urban Area of Barrackpore Municipality.*" Manuscript in preparation, 2026.

Manuscript in
Preparation (2026)

Co-authors, **Saptarshi Sadhu** (4th Author). "*Enhancing Micro Electrochemical Spark Machining of Exotic Non-Ferrous Materials through On-Process Electrolyte Sonication.*" Manuscript in preparation, 2026.

Open Source Leadership

Mar 2026 – Pres.

Lead Maintainer & Systems Architect

EaseMotion CSS Framework

[GitHub Repository](#).

- **Global & National Contributor Rank:** Ranked **#3 top private committer in India** ([Committers.top Private](#)) and **#2 active in commits** ([Committers.top Public](#)).
- **Large-Scale Project Maintenance:** Orchestrated codebase evolution under GirlScript Summer of Code (GSSoC) 2026, scaling framework to **21,587 merged pull requests, 54,166 commits, and 642 contributors** in 50 days (avg 496 PRs/day).
- **Automated Anti-Spam Architecture:** Deployed a **Honeypot Sandbox Relocation System** to automatically route, isolate, and audit point-farming spam folders.
- **CI/CD Optimization:** Engineered GitHub Actions pipelines with caching and REST/GraphQL API optimizations, cutting validation runtime from **3 minutes to under 25 seconds** per PR.

Technical Projects

Full-Stack	CivicIssue Dashboard GitHub Repository MVC Architecture. Developed a full-stack MVC application integrating bcrypt password hashing, session management, input validation, and protected routing for city-level civic complaint reporting.
JavaScript	Multi-Source Weather Forecast App GitHub Repository REST API & Async JS. Built an asynchronous weather forecasting tool featuring a backend proxy server to prevent API rate-limiting, custom caching layers, and automated fallback endpoints.
AI Platform	Techtonics – AI Assisted Coding Platform GitHub Repository AI API Integration. Developed an interactive learning platform connecting real-time AI tutor APIs to guide beginners with step-by-step logic breakdown and syntax diagnostics.
Web App	Personal Portfolio Website GitHub Repository Responsive Web. Designed and developed a personal portfolio showcasing projects and technical skills with responsive design across devices, integrated project links, and optimized SEO.

Technical Skills

Programming	Python, C, C++, Java, JavaScript, SQL, LaTeX
Machine Learning	Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Model Evaluation, Regression, Classification, Time Series, Explainability (SHAP)
Web	React.js, Node.js, Express.js, HTML5, CSS3, Tailwind CSS, Bootstrap, EJS, REST APIs
Databases & DevOps	MongoDB, MySQL, Supabase, Git, GitHub Actions, CI/CD Automation, Linux
GIS	Google Earth Engine, QGIS, Raster Analysis, NDVI, LST, Satellite Analytics

Awards & Achievements

July 2026	ACM Summer School at IISC Bangalore: Selected to attend the summer school at IISC Bangalore
May 2026	Google Student Ambassador (2026): Selected campus representative to lead technical programs and AI workshops.
April 2026	4th Place Coding Premier League: Algorithmic problem-solving finishes.
February 2026	AWS College Showcase: Selected among top 7 university teams for cloud innovation.
June - July 2025	LetsUpgrade: Campus Ambassador
Mar 2025	Clash of Coders 2.0 Finalist : Algorithmic problem-solving finishes and Software making brainstorming competition.
HackerRank	4-Star Rating in C and 3-Star Rating in C++ / Problem Solving on HackerRank platform.

Leadership & Service

- Apr 2026 – Pres.** **Publicity Chair**
ACM Student Chapter, Adamas University
Kolkata, India.
Architected promotional campaigns driving 30% growth in chapter attendance, managed social media channels, and coordinated technical guest lectures.
- Mar 2026 – Pres.** **Student Coordinator**
IETE Student Chapter, Adamas University
Kolkata, India.
Spearheaded student events, maintained membership databases, and served as primary student-faculty liaison.
- Oct 2025 – Pres.** **Class Representative (CR)**
Department of CSE, Adamas University
Kolkata, India.
Advocated academic resources and resolved concerns for 60+ computer science undergraduates.
- Feb 2026** **Team Leader**
AI Impact Summit Buildathon (HCL GUVI)
Kolkata, India.
Led development of a multilingual AI vs. human text detection system across 5 languages, maintaining 6+ days continuous uptime via ngrok.
- Sep 2024 & Sep 2025** **Team Leader**
Smart India Hackathon (SIH) 2024 & 2025
Adamas University, Kolkata.
India
Directed project timelines, brainstorming sessions, and solution design across two SIH editions.

Academic Exposure

- Quantum Lab, 16th April 2026** **TCG CREST Quantum Lab Visit (CQuERE):** Academic visit exploring quantum computing algorithms and experimental research.
- Data Centre, 16th October 2025** **CTRLS Enterprise Data Centre Visit:** Industrial visit gaining direct exposure to enterprise server infrastructure and cloud architecture.

Certifications

- 2026** **AI Impact Summit Buildathon Certificate** — HCL GUVI
- 2025** **Urban Vision Hackathon 2025 Certificate of Achievement** — Ministry of Education, Govt. of India, Organised by IISC Bangalore
- 2025** **Full Stack Web Development Training** — DevTown
- 2025** **Data Structures & Algorithms using C++** — DevTown
- 2025** **Competitive Programming Course** — DevTown

Languages

- Proficiency** **English** (Upper Intermediate Proficient, B2) · **Bengali** (Native) · **Hindi** (Proficient, C2) · **German** (Beginner, A1)